



Elaborato finale per il conseguimento  
della Laurea in Ingegneria aerospaziale

## Analysis of Direct Earth-Moon Transfers

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**Abstract.** The lunar exploration has always been an important challenge for humankind. Several missions have been carried out, but the interest in our moon remains strong. This thesis analyzes direct transfers between the Earth and the Moon, under the assumption of a planar motion and using the patched conics approximation. To fully achieve the mission objectives and articulate the practical significance of this type of transfer, detailed descriptions are provided for two potential outcomes: achieving orbital capture and executing a fly-by maneuver. Both study cases are examined in terms of  $\Delta V$  acquired or used, exploring how adjustments in the initial angle impact the results. The analytical findings are verified by numerical integration of the motion equations using MATLAB, employing a patched conics approach, and GMAT, which incorporates a more realistic dynamical model

### 1. Introduction

Aside from Earth, the Moon is the celestial body for which we have the most comprehensive knowledge. However, it still harbors countless mysteries waiting to be uncovered about its formation, geological history, and potential resources [1]. From a scientific perspective, lunar exploration has always been relevant due to the possibility to study the story of our solar system through surface analysis and to carry on astronomical observations [2]. Since 1959, when the first spacecraft reached the Moon, the USSR succeeded in several missions with increasingly higher imag-

ing quality. Similarly, from 1961, the Americans began ramping up their efforts through the Apollo program. These missions were initially designed to impact the surface while taking pictures. In 1969 the Apollo 11 brought the first human on the Moon's surface. Human missions continued until 1976, after which there was a 20-year gap in lunar exploration [3]. However, the past decade has witnessed a renaissance in lunar exploration, driven by new objectives such as establishing long-term human presence on the Moon and reducing exploration costs [4]. These necessities are the driving force behind this thesis, which focuses on designing direct transfers to the Moon by thoroughly



evaluating all necessary parameters. The study aims to ensure not only the success of the transfer but also the subsequent observation phase. To achieve this, the analysis focus on the orbital capture, comparing various parameters for different periapsis altitudes to determine the most efficient and effective approaches. Furthermore, an additional discussion explores the potential advantages of utilizing a lunar flyby for gravity assist maneuvers. This section considers multiple scenarios to evaluate how a flyby can be strategically employed to optimize the mission's trajectory and reduce fuel consumption. By examining these aspects, the thesis provides a comprehensive framework for future lunar missions, aiming to enhance our ability to reach and study the Moon efficiently.

## 2. Problem Statement

The objective is to design a preliminary trajectory for transferring between the Earth and the Moon, using approximations that are acceptable for this type of mission. To keep the problem planar, the spacecraft departs from a circular orbit around Earth that lies in the same plane as the Moon's orbit. The pericenter of the transfer orbit has radius  $r_0$  and velocity  $V_0$  assumed given throughout this work. All maneuvers are considered to be impulsive. The problem is solved using the patched conics approximation, which introduces the concept of the sphere of influence (SOI), a region where the gravitational pull of a central body is dominant with respect to all the others. Consequently, the analysis is divided into two phases. The first phase concerns the transfer orbit, which occurs entirely within the Earth's SOI, the equations of motion are as follows

$$\ddot{\mathbf{r}} = -\frac{\mu_E}{r^3} \mathbf{r} \quad (1)$$

where  $\mu_E$  is Earth's gravitational parameter and  $\mathbf{r}$  is the position vector of a body relative to the Earth. Therefore, it is simplified as a two-body problem, involving only the Earth and the spacecraft, without accounting for any perturbations. This phase concludes when the spacecraft intercepts the Moon's sphere of influence. The second phase, governed by the Moon's gravity, encompasses either the or-

bital capture or the fly-by maneuver. The motion equations consider the same approximation and are the following

$$\ddot{\mathbf{g}} = -\frac{\mu_M}{g^3} \mathbf{g} \quad (2)$$

where  $\mathbf{g}$  is the position vector of a body relative to the Moon and  $\mu_M$  is the Moon's gravitational parameter.

## 3. Analytical Solution

It is important to define the reference frame used for this part of the mission, which will be an Object Referenced type. The x-axis will point towards the Moon at the time of interception with the Moon's SOI. The z-axis will be parallel to the angular momentum vector of the Moon's orbit, and the y-axis completes a right-handed coordinate system. The SOI radius is

$$\rho_{SOI} = \left( \frac{\mu_M}{\mu_E} \right)^{\frac{2}{5}} L \quad (3)$$

where  $L$  is the Earth-Moon distance. For the problem at hand one has to determine the optimal phasing at the departure, when the first maneuver occurs. The parameter that need to be set is the angle between the vector pointing to the Moon from Earth and the vector between Earth and the SOI interception point  $\alpha_2$ , as shown in Fig. 1. The remaining parameters of the transfer orbit are easily evaluated once  $\alpha_2$  is known. The semi-major axis is obtained as

$$a_1 = -\frac{\mu_E}{2\left(\frac{v_0^2}{2} - \frac{\mu_E}{r_0}\right)} \quad (4)$$

Instead, the eccentricity can be evaluated from the angular momentum, which is  $h_1 = r_0 v_0$  as the velocity  $v_0$  is orthogonal to the periapsis radius  $r_0$ , with the following equation

$$e_1 = \sqrt{1 - \frac{h_1^2}{\mu_E a_1}} \quad (5)$$

The evaluation of the true anomaly at the encounter with the Moon's SOI  $\nu_1$ , is then needed to determine  $\alpha_1$ , which is the angle between the x-axis and the direction of the periapsis, represented in Fig. 1 by the eccentricity vector.

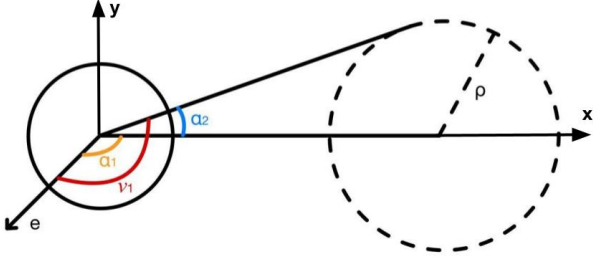


Figure 1: Starting angles.

Therefore, the angle between the x-axis and the starting position is  $\alpha_1 = \nu_1 - \alpha_2$ . To ensure interception with the SOI,  $\alpha_2$  must satisfy the following relation

$$|\alpha_2| < \arctan\left(\frac{\rho}{L}\right) \quad (6)$$

After choosing a suitable parameter, it is easy to evaluate the remaining angles using geometric relations on the triangle shown in the Fig. 2.

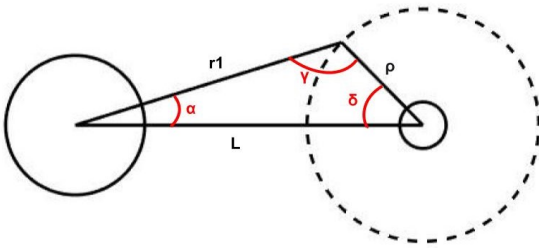


Figure 2: evaluation of  $r_1$ .

In particular, using the law of sines it is possible to get the angle  $\gamma$

$$\frac{\rho}{\sin \alpha_2} = \frac{L}{\sin \gamma} \Rightarrow \gamma = \arcsin\left(\frac{L}{\rho} \sin \alpha_2\right) \quad (7)$$

from which it is easy to calculate the distance  $r_1$ .

$$r_1 = \rho \frac{\sin \delta}{\sin \alpha_2}$$

where the relationship  $\delta = \pi - \gamma - \alpha_2$  has been exploited. The true anomaly  $\nu_1$  is obtained as

$$\nu_1 = \arccos\left(\frac{1}{e_1} \left(\frac{a_1(1 - e_1^2)}{r_1} - 1\right)\right) \quad (8)$$

The transfer time length is evaluated using Kepler law

$$\Delta t_{or} = \sqrt{\frac{a_1^3}{\mu_E}} (E - e_1 \sin E) \quad (9)$$

where

$$E = 2 \arctan\left(\sqrt{\frac{1 - e_1}{1 + e_1}} \tan\left(\frac{\nu_1}{2}\right)\right) \quad (10)$$

is the eccentric anomaly. To perform the next phase, the velocity at the end of the transfer  $V_1$  is needed, which can be obtained through the vis-viva equation as follows

$$V_1 = \sqrt{\mu_E \left(\frac{2}{r_1} - \frac{1}{a_1}\right)} \quad (11)$$

### 3.1. FlyBy

The parameters that will be evaluated to perform the flyby need to be expressed in a frame centered on the Moon. Specifically, a hyperbolic frame was chosen. This frame is composed of a ternary  $[u_r, u_v, u_h]$ , where the second component is aligned with the Moon's velocity, the third component is orthogonal to the flyby plane, and the first component completes a right-handed coordinate system. To obtain the state vector in the new frame, two angles are needed: the angle between the spacecraft's velocity and the Moon's velocity ( $\eta$ ), and the angle between  $\hat{u}_r$  and the vector pointing from the Moon to the spacecraft ( $\epsilon$ ). The first step for each of these two evaluations is to determine the inclination ( $\psi$ ) of the Moon's velocity relative to the y-axis of the frame used for the transfer orbit. This angle is the flight path angle of the moon and it's easily to get through the following formula:

$$\psi = \arctan\left(\frac{e_m \sin \nu_m}{1 + e_m \cos \nu_m}\right) \quad (12)$$



The same equation is then used for the spacecraft's flight path angle with respect to the Earth,  $\gamma_e$ . The angle  $\epsilon$  is now evaluated by geometric relations on the triangle represented in Fig. 3

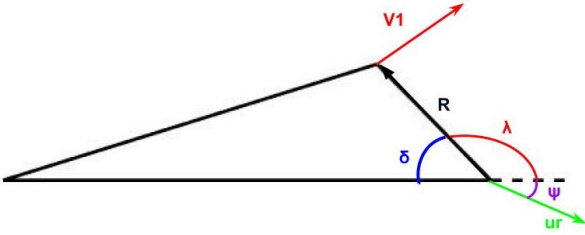


Figure 3:  $\epsilon$  evaluation.

where  $R$  is the position vector of the spacecraft with respect to the moon and its magnitude is exactly  $\rho$ .  $\delta$  is already known from previous analysis, so with the following equations it can be derived  $\lambda$  and then  $\epsilon$ :

$$\lambda = \pi - \delta \quad \epsilon = \lambda + \psi \quad (13)$$

From  $\gamma_e$  it can be obtain the angle between the spacecraft's velocity and the direction orthogonal to the distance vector from Earth to the Moon. It is done with geometric relations with respect to Fig. 4

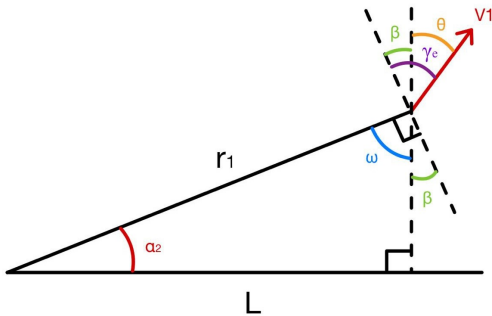


Figure 4:  $\eta$  evaluation.

The two dashed line are orthogonal respectively to  $r_1$  and  $L$ , the angle needed is  $\theta$  which can be obtained by the following relations

$$\omega = \frac{\pi}{2} - \alpha_2 \Rightarrow \beta = \frac{\pi}{2} - \omega = \alpha_2 \quad (14)$$

$$\theta = \gamma_e - \beta = \gamma_e - \alpha_2 \quad (15)$$

$\theta$  is not exactly  $\eta$  because the moon velocity is not orthogonal to its radius due to the eccentricity of moon's orbit. Therefore the following correction is needed

$$\eta = \theta - \psi \quad (16)$$

Now the state vector could be expressed in the hyperbolic frame and the flyby can be entirely described. Specifically the velocity  $V_1$  is written as a vector as follow

$$V_1 = V_1 \cos \eta u_v + V_1 \sin \eta u_r \quad (17)$$

The first element needed is the hyperbolic excess velocity of the spacecraft, which is  $v_{\infty 1} = V_1 - V_M$ , where  $V_M$  is the velocity of the Moon, obtained using the vis-viva equation and expressed as a vector (having only a component along  $\hat{u}_v$ ). Using  $\epsilon$  it is possible to define the position vector in the hyperbolic frame. Finally, it is possible to get all the orbital elements of the orbit; however the only relevant parameters are the semi major axis and the eccentricity, which are used to evaluate the declination ( $\delta_2$ ) and the velocity at the end of the fly-by. Therefore, using the same procedure shown in the first section, it is estimated  $e_2$  and  $a_2$  and then  $\delta_2$  with the following equation

$$\delta_2 = 2 \arcsin \left( \frac{1}{e_2} \right) \quad (18)$$

Obtaining the hyperbolic excess velocity at the end of the flyby is straightforward using a rotation matrix with an angle  $\delta_2$  along the z-axis as follows

$$v_{\infty 2} = R_z(\delta_2) v_{\infty 1} \quad (19)$$

Although, the velocity of the spacecraft at the exit of the SOI  $V_2$ , is obtained by adding to  $v_{\infty 2}$  the moon's velocity at the end of the flyby. To get this velocity, the duration time of this phase of the mission should be calculated, this is equal to two times the seconds to reach the pericenter

$$t_{flyby} = 2 \sqrt{-\frac{a_2^3}{\mu_M}} (e_2 \sinh F - F) \quad (20)$$



where  $F$  is the hyperbolic anomaly

$$F = \operatorname{arcosh} \left( \frac{a_2 - \rho}{a_2 e_2} \right) \quad (21)$$

Now, the velocity of the Moon after  $t_{flyby}$  can be easily obtained, and consequently, the velocity  $V_2$  can also be determined. The difference in magnitude between  $V_2$  and  $V_1$  quantifies the acceleration or deceleration resulting from the gravity assist.

### 3.2. Orbital Capture

The other possible outcome of the mission is orbital capture, specifically into a circular orbit. The procedure follows the same steps as described for the flyby; however, this time an impulsive burn will be performed at periselene to reduce velocity and insert the spacecraft into its final orbit. This maneuver is executed at periapsis, where the velocity is orthogonal to the radius and it can indeed be performed exclusively in the direction of the velocity. The only necessary parameter is the perifocal distance from the Moon, which is straightforward to calculate since  $a_2$  and  $e_2$  are known from previous analyses. The relation used is the following

$$g_2 = a_2(1 - e_2) \quad (22)$$

It is therefore important to compute the capture  $\Delta V$ . This value represents the difference between the velocity at periapsis, calculated using the vis-viva equation, and the velocity of the circular orbit, obtained using the following relation

$$V_{c2} = \sqrt{\frac{\mu_M}{g_2}} \quad (23)$$

## 4. Computational approach using MATLAB

By employing a function like "ode113" it is possible to integrate the motion equation in MATLAB to verify the obtained results. The transfer orbit takes place entirely within the Earth's SOI, so the gravity model under consideration pertains solely to the central body. The information needed for the integration are the transfer time and the initial state

vector, both computed previously. The results are plotted in Fig. 5

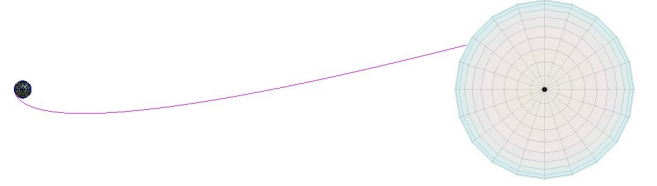


Figure 5: Transfer orbit.

The orbit obtained matches the one computed through the analytic approach, considering that the angle  $\alpha_2$  was set to  $4.010^\circ$ . Proceeding in the same manner as for the transfer orbit, it is possible to propagate the flyby trajectory and the orbital capture through MATLAB. The same motion equations are used considering the moon as the central body, as the spacecraft is within the Moon's sphere of influence. The initial state has already been computed since  $V_2$  and  $R$  are known, as well as the velocity  $V_{c2}$  and the distance  $g_2$  for the circular orbit. The results, plotted in Fig. 6, are exactly

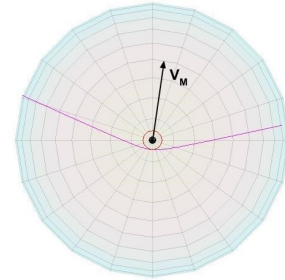


Figure 6: Fly-by and orbital capture.

the one expected from the analytical approach. The flyby obtained is indeed on the trailing side, as expected, and the capture begins exactly at the periapsis, which has the expected altitude from the surface.

## 5. Computational approach using GMAT

To have a more precise verification, GMAT [5], a numerical solver, is considered. For the sake of comparison, the ini-



tial state vector is defined in the same reference frame as before. However, the vector pointing to the Moon will differ because it is referenced to the starting epoch. To obtain correct results, the state vector is rotated by an angle between the positions of the Moon at the two different epochs in the orbital plane frame. This angle is easily obtained by multiplying the transfer time derived before to the mean motion of the moon

$$n_M = \sqrt{\frac{\mu_M}{(a_M)^3}} \quad (24)$$

The dynamic model used resembles the previous analysis, with the orbit transfer propagated considering Earth as the primary body. The key difference from the numerical results obtained in MATLAB are due to the presence of various perturbations. Specifically, throughout the entire transfer orbit, perturbations caused by the Sun and the Moon, as well as the radiation pressure from the Sun, are taken into account. As expected the orbit obtain, shown in Fig. 7, closely resembles the one obtained through MATLAB.

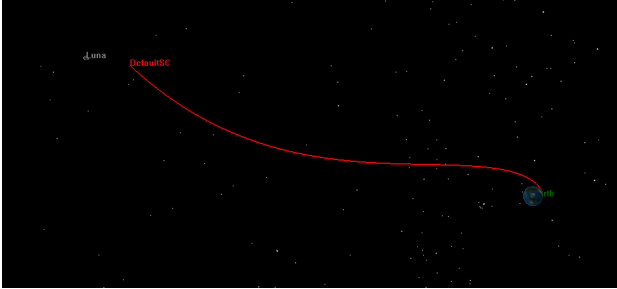


Figure 7: Transfer orbit GMAT.

## 6. Numerical results

In this section, several numerical values are presented for a detailed analysis of the mission design. A preliminary analysis was conducted with a starting velocity of  $V_0 = 10.895 \text{ km/s}$  and a starting position of  $r_0 = 6700 \text{ km}$ , producing an elliptical transfer orbit ( $e_1 = 0.995$ ). For the analytical evaluation,  $\alpha_2$  was set to  $4.010^\circ$ , resulting in a starting angle of  $\alpha_1 = 160.8^\circ$ . This angle was then used for propagation in MATLAB and GMAT, producing the following

results at the end of the transfer orbit, where the distance from the moon is exactly  $\rho$

Table 1: Transfer orbit's parameters.

| Parameters      | Analytic        | Matlab          | GMAT            |
|-----------------|-----------------|-----------------|-----------------|
| $\nu_1$         | $164.884^\circ$ | $164.885^\circ$ | $141.604^\circ$ |
| $\Delta t_{or}$ | 44.1860 h       | 44.1864 h       | 42.4214 h       |
| $ V_1 $         | 1.433 18 km/s   | 1.433 19 km/s   | 1.632 42 km/s   |

The parameters under consideration are crucial for the subsequent phases of the mission. The difference in percentage between the analytical approach and MATLAB is approximately 0.001 % for each parameter. In contrast, the variations observed with GMAT propagation are much more pronounced, with an average deviation of about 12 %, which is likely due to perturbation effects.

### 6.1. Flyby

Using the results obtained from the analytical approach with the same  $\alpha_2$  as before, the fly-by can be entirely described by the following parameters

Table 2: Flyby parameters.

| Parameter   | Value                   |
|-------------|-------------------------|
| $a_2$       | $-2074.72 \text{ km}$   |
| $e_2$       | 3.27943                 |
| $\delta_2$  | $35.5081^\circ$         |
| $V_2$       | $1.964 70 \text{ km/s}$ |
| $t_{flyby}$ | 23.5958 h               |
| $\Delta V$  | $0.531 52 \text{ km/s}$ |

The obtained  $\Delta V$  indicates that the flyby occurs on the trailing side, causing the spacecraft to accelerate. An interesting study could be conducted to observe the changes in acceleration caused by the gravity assist by varying the angle  $\alpha_2$ . The optimal condition is achieved when  $\alpha_2 = 4.436^\circ$ , resulting in a  $\Delta V$  of  $837.48 \text{ m/s}$ . This represents the highest velocity change, not considering situations of leading



side which would cause deceleration.

## 6.2. Orbital capture

Keeping the same  $\alpha_2$  used in the last two section, the parameters obtained for the orbital capture phase are tabulated in Table 3. The orbit's altitude obtained does not meet

Table 3: Orbital capture parameters.

| Parameter  | Value      |
|------------|------------|
| $g_2$      | 4729.17 km |
| $V_{c2}$   | 1.018 km/s |
| $\Delta V$ | 1.088 km/s |

specific requirements; however, for a real mission, certain constraints are necessary for observational purposes. It is valuable to analyze how variations in  $\alpha_2$  correlate with different periapsis altitudes and, particularly, with changes in  $\Delta V$ . The results are depicted in Fig 8.

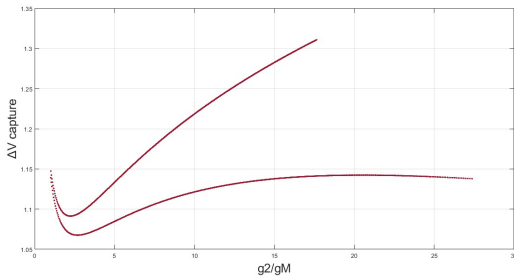


Figure 8: Analysis of  $\Delta V$  for capture.

The graph illustrates that the same distance can be achieved with two different  $\Delta V$  values derived from distinct  $\alpha_2$  angles. The orbit requiring the lowest  $\Delta V$  has a perifocal distance of  $2.6 g_M$ , where  $g_M$  is the lunar radius.

## 7. Conclusions

A comprehensive mission plan has been proposed, and the final orbit achieved via direct transfer has met initial considerations and has been validated through various ap-

proaches, yielding positive outcomes. Regarding the fly-bys, it has been demonstrated that modest acceleration can be achieved through lunar gravity, which holds significant promise for real interplanetary missions. The orbital capture phase confirms the ability to attain different altitudes while offering an alternative approach to reduce fuel costs. Future developments should concentrate on analyzing various types of direct transfers comparing them with the designed orbit. This analysis aims to fully understand the potential for realizing such missions, highlighting the possible changes needed to increase reliability.

## References

- [1] National Research Council, Division on Engineering, Physical Sciences, Space Studies Board, and Committee on the Scientific Context for Exploration of the Moon. *The scientific context for exploration of the Moon*. National Academies Press, 2007.
- [2] Paul Lucey, Randy L Korotev, Jeffrey J Gillis, Larry A Taylor, David Lawrence, Bruce A Campbell, Rick Elphic, Bill Feldman, Lon L Hood, Donald Hunten, et al. Understanding the lunar surface and space-moon interactions. *Reviews in mineralogy and geochemistry*, 60(1):83–219, 2006.
- [3] Ian A Crawford and Katherine H Joy. Lunar exploration: opening a window into the history and evolution of the inner solar system. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 372(2024):20130315, 2014.
- [4] Peng Zhang, Wei Dai, Ran Niu, Guang Zhang, Guanghui Liu, Xin Liu, Zheng Bo, Zhi Wang, Haibo Zheng, Chengbao Liu, et al. Overview of the lunar in situ resource utilization techniques for future lunar missions. *Space: Science & Technology*, 3:0037, 2023.
- [5] NASA's Technology Transfer Program. General mission analysis tool (gmat) v.r2016a. <https://software.nasa.gov/software/GSC-17177-1>, 2016. Accessed: March 13, 2024.